

# Data Cleaning & Preparation - Change Log

Project 1 | Dataset: Dataset\_for\_Data\_Analytics.xlsx | 1,200 order records | DecodeLabs Industrial Training Kit

## Summary

A full audit of the raw dataset was performed covering duplicate detection, missing-value handling, date/number format standardization, text normalization, and cross-field business-rule validation. The table below documents every check performed and any corrective action taken, per the stakeholder accountability requirement (what changed and why).

| Change ID | Description                                                              | Impact                                                                                                                  | Status   |
|-----------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------|
| CR001     | Audited for duplicate rows and duplicate OrderIDs                        | Found 0 duplicate rows, 0 duplicate OrderIDs (dataset was already unique)                                               | Verified |
| CR002     | Imputed missing 'CouponCode' values                                      | 309 blank records (orders with no coupon applied) set to "No Coupon" for consistency                                    | Resolved |
| CR003     | Corrected 'TotalPrice' numeric precision                                 | 29 records had floating-point rounding artifacts beyond 2 decimals; rounded to standard currency precision (2 decimals) | Resolved |
| CR004     | Standardized 'Date' field to ISO 8601 (YYYY-MM-DD)                       | 0 unparseable dates found; all dates normalized to ISO 8601 text format for consistency                                 | Resolved |
| CR005     | Trimmed whitespace and verified text casing on all categorical/ID fields | 0 cell(s) had leading/trailing whitespace requiring trimming; casing was already consistent across all category values  | Verified |
| CR006     | Validated 'TotalPrice' = Quantity x UnitPrice business rule              | 0 records failed the cross-field validation check                                                                       | Verified |
| CR007     | Checked for negative or zero values in Quantity, UnitPrice, TotalPrice   | 0 invalid (negative/zero) numeric records found                                                                         | Verified |
| CR008     | Final verification gate check                                            | Duplicate ID error rate: 0.0%   Incorrect date format error rate: 0.0%                                                  | Passed   |

## Verification Gate (Project 2 Threshold)

| Check                       | Result            | Status |
|-----------------------------|-------------------|--------|
| Duplicate Order IDs         | 0 of 1,200 (0.0%) | Pass   |
| Incorrectly formatted dates | 0 of 1,200 (0.0%) | Pass   |

Both zero-error thresholds required to advance to Project 2 have been met.

## Methodology Notes

- Missing values: Listwise deletion was avoided per best practice. The only field with nulls was CouponCode (309 records), which represents orders where no coupon was applied - these were labeled 'No Coupon' rather than dropped, preserving all 1,200 records.
- Duplicates: Checked both for fully duplicated rows and duplicate unique identifiers (OrderID, TrackingNumber). None were found; the source data was already unique.
- Formats: Date field standardized to ISO 8601 (YYYY-MM-DD). Currency fields (TotalPrice) rounded to 2 decimal places to remove floating-point artifacts (29 records affected). Text fields checked for whitespace and casing inconsistencies.
- Business rules: Cross-validated TotalPrice against Quantity x UnitPrice, and Quantity against ItemsInCart, to confirm internal consistency of every record.